

- 3 (a) Fig. 3.1 is a photomicrograph of a section through two different types of blood vessels, **X** and **Y**.

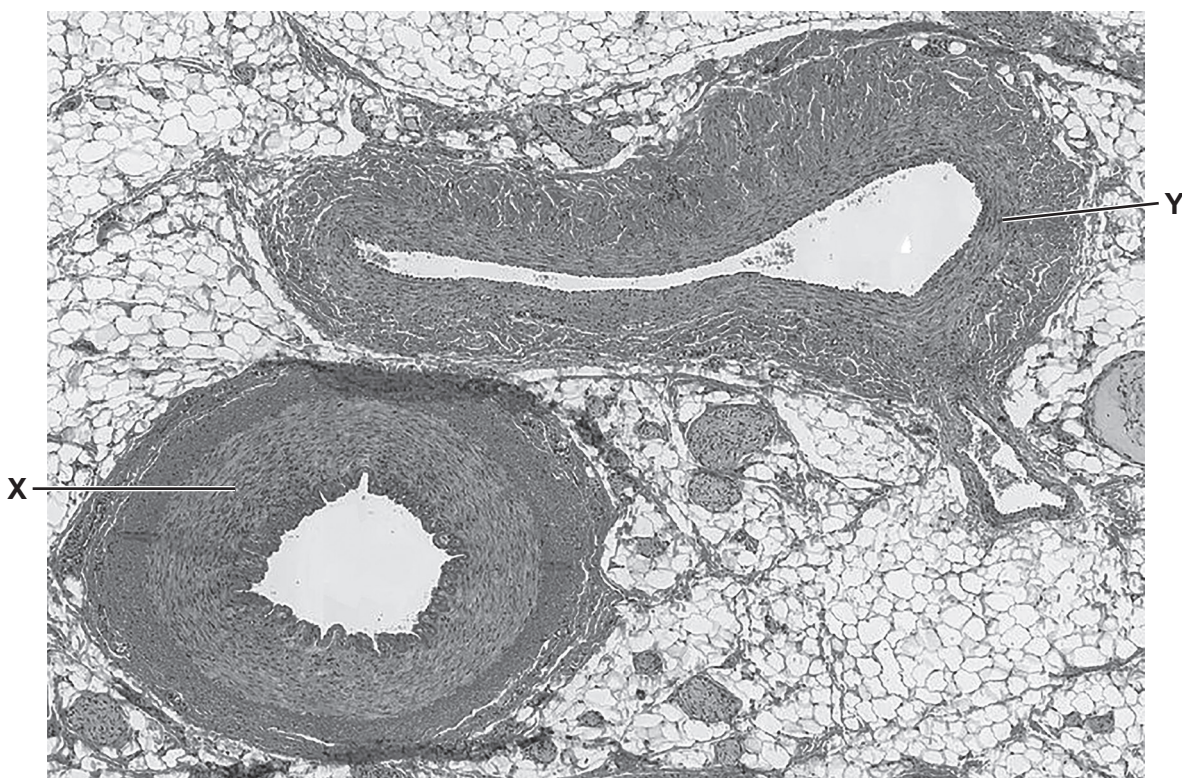


Fig. 3.1

- (i) Name the two types of blood vessel shown by **X** and **Y** in Fig. 3.1.

X

Y

- (ii) State the reasons for your identification of the type of blood vessel shown by Y in Fig. 3.1.

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- (b) Tissue fluid and lymph are formed when blood arrives in the capillary networks of body tissues.

- (i) Explain why tissue fluid is more similar to blood plasma than it is to blood.

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- (ii) Explain why the tissue fluid formed after blood arrives in the capillary network has a **higher** concentration of amino acids than the newly formed lymph draining away from the network.

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- (c) The trachea, bronchi and bronchioles in the gas exchange system require a supply of glucose and oxygen from the blood for the functioning of smooth muscle.

Outline the function of smooth muscle in the gas exchange system.

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[Total: 8]